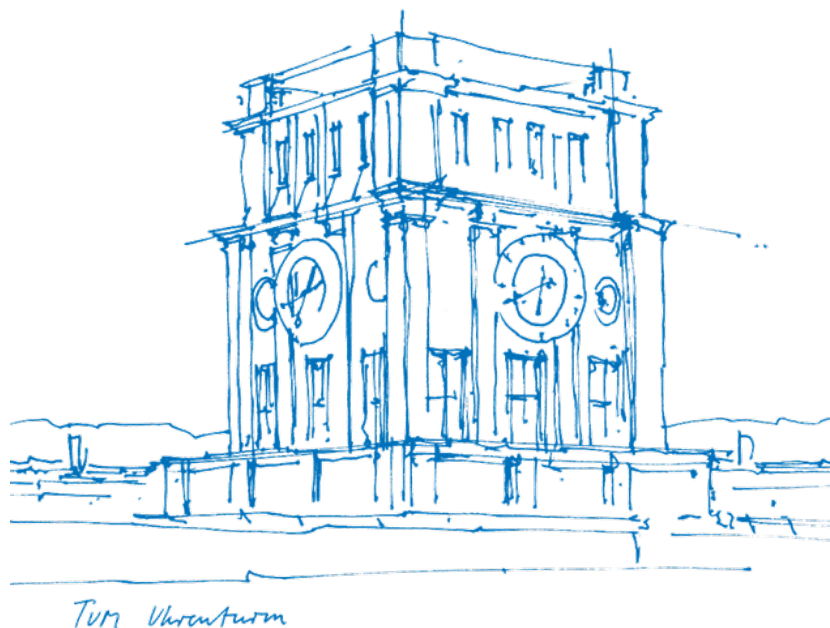


Master's Thesis in Information Systems

Xin Wen

Document-Based Process Modeler



Master's Thesis in Information Systems

Xin Wen

Document-Based Process Modeler

Dokumentatenbasierte Bearbeitung und Änderung von
Prozessmodellen

Thesis for the Attainment of the Degree
Master of Science

at the TUM School of Computation, Information and Technology,
Department of Computer Science,
Chair of Information Systems and Business Process Management (i17)

Examiner

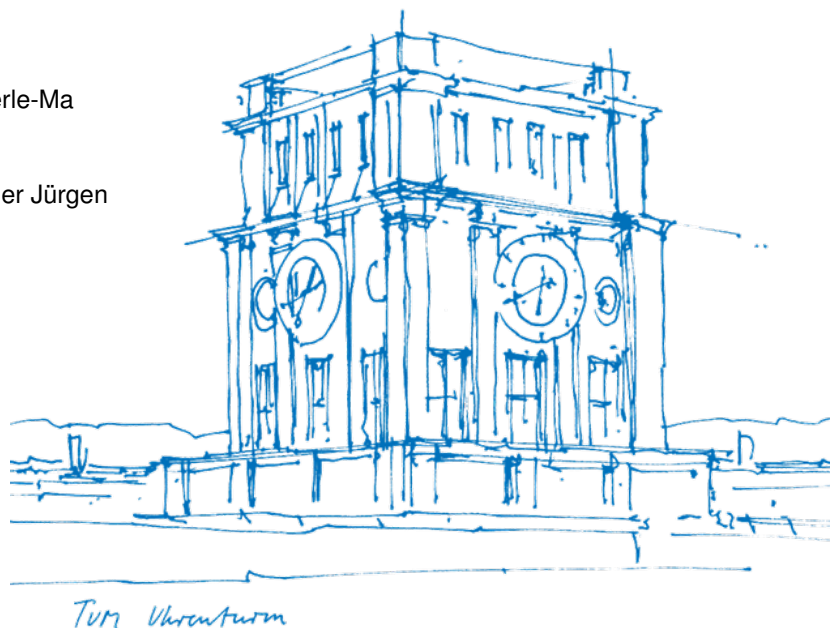
Prof. Dr. Stefanie Rinderle-Ma

Supervised by

Dr. rer. soc. oec. Mangler Jürgen

Submitted on

15.04.2026



Declaration of Academic Integrity

I confirm that this master's thesis is my own work and I have documented all sources and material used.

I am aware that the thesis in digital form can be examined for the use of unauthorized aid and in order to determine whether the thesis as a whole or parts incorporated in it may be deemed as plagiarism. For the comparison of my work with existing sources I agree that it shall be entered in a database where it shall also remain after examination, to enable comparison with future theses submitted. Further rights of reproduction and usage, however, are not granted here.

This thesis was not previously presented to another examination board and has not been published.

Garching, 15.04.2026

Xin Wen

Abstract

Keywords: *Process Modeling*

Contents

1	Introduction	7
1.1	Motivation	7
1.2	Research Questions	7
1.3	Contribution	8
1.4	Methodology	8
1.5	Evaluation	8
1.6	Structure	9
2	Related Work	9
2.1	Text2Model Transformation? Text-based Process Modeling?	9
2.2	LLM-based Process Modeling Tool	9
	ProMoAI	9
	BPMN-Chatbot	9
	AutoBPMN.AI	9
2.3	Document AI Tool	9
	NotebookLM?	9
2.4	Human-in-the-loop	10
3	Solution Design	10
3.1	Requirements	10
	Functional Requirements	10
	Non-functional Requirements	10
3.2	Design Overview	10
	Views	11
3.3	Modeling Workflow Design	12
3.4	Core Functionalities Design	12
	Document Management	12
	Document Interaction	12
	Process modeling	12
	Traceability	13

	5
Project Graph Visualization	13
3.5 Interaction Design	14
3.6 System Architecture	14
3.7 Conceptual Data Model	16
4 Implementation	18
4.1 Frontend	18
Architecture	18
4.2 State Management	18
Document Processing	19
Text Selection Rendering	19
Modeling	19
Rendering Customization	19
Architecture	19
4.3 Backend	19
Architecture	19
4.4 Database	20
5 Evaluation	20
5.1 Evaluation Procedure	20
5.2 Evaluation Results	21
Overall Perception	21
Positive Aspects	24
Challenges in Interaction and Use	24
Potential Improvements and Additional Features	24
5.3 Additional Observations	24
6 Discussion	25
6.1 Usability and User Experience	25
6.2 User Expectations	25
7 Conclusion	25
7.1 Summary	25

	6
7.2 Future Work	25
A Appendix	26
A.1 Evaluation Material	26
B2B Order Processing Scenario	26

List of Tables

1 Summary of Evaluation Results by Functionality	23
--	----

List of Figures

1 Workspace UI Layout	11
2 Syntax example of the augmented model data.	13
3 System Architecture	15
4 Conceptual Data Model	16
5 Frontend codebase structure	18
6 Example JSON schema used in the backend for text quotes	19
7 Backend Component	20
8 Modeling result	22
9 Modeling result	22
10 Different modeling strategies	24

Introduction

Motivation

Business process is a valuable asset of an organization. But most of them are stored in unstructured format, e.g., Word documents, PDF files. Manual transformation from unstructured text to structured process models is time-consuming and error-prone.

In the era of Gen-AI, LLM has proved to be a promising solution to support creating process modeling efficiently. However, on one hand it still cannot fully replace human in process modeling especially when domain knowledge is needed. On the other hand, existing LLM-based process modeling tools and approaches still have limitations especially in the scenario for creating process models from documents. For example, the typical "one view one model" approach make it hard to manage for large systems, where multiple process and subprocess models are needed. In addition, the usual separate interface of the document and the generated model make it hard to verify the result. Besides, user need to manually extract relevant information from documents and input it into the conversational chatbot.

To address these limitations, this work proposes an integrated tool that supports process modeling from documents with a more seamless, efficient, and user-friendly workflow by offering the following core features: (1) manage all the related documents and models for one use case within one unified view (2) allowing users to upload documents and select passages from the documents to serve as input for LLM-based process model generation directly, (3) providing a clear visual link between the text and the generated model through display the document and the model side by side

Research Questions

This work addresses the following research questions:

RQ1 How do existing approaches and tools support process modeling from documents?

This question investigates how existing approaches and tools support process modeling from documents by identifying their functionalities and capabilities. It helps guide the design of the proposed system.

RQ2 How can an integrated system be designed to support process modeling from documents?

This question addresses the design of the proposed system. It focuses on what functionalities should be integrated and how to integrate them, aiming to enable a seamless modeling workflow with high usability.

RQ3 How do users perceive the proposed system?

This question evaluates the proposed system from the users' perspective. It aims at understanding how users perceive the system's benefits and limitations. Its results provide insights into the system's strengths and weaknesses and help identify potential design issues and areas for improvement.

Contribution

The proposed and developed novel tool as an artefact of this work that provides a series of functionalities to support process modeling from documents in an innovative way: - It allows users to manage with multiple documents and multiple models for one use case within one unified interface. - It allows a new input method for LLM-based process model generation by allowing users to select multiple passages from one document to serve as input for LLM-based process model auto-generation.

Methodology

This work follows the three interconnected cycles of DSR **Hevner2007** proposed by Hevner: The relevance cycle: the first cycle is done through the communication with modeling experts, who proposes the tool and elicitate the requirements in section 3.1 for the tool. The rigor cycle: it is done through the review in section 2. The design cycle: the first cycle is done through the delivered solution of RQ2 and evaluation result of RQ3. Through the analysis and discussion of the evaluation result, we can identify the strengths and weaknesses of the proposed tool, and further improve the design of the tool in the next iteration.

Evaluation

The evaluation of the proposed system is done with 4 experts in qualitative way. It consisted of task-based activities and a follow-up questionnaire to collect user feedback on the tool's usability and perceived usefulness.

Structure

The remainder of this thesis is structured as follows: Chapter 2 reviews the related work of LLM-based process modeling tools. Chapter 3 presents the design solution of the tool, including its requirements, architecture, and functionalities. Chapter 4 describes its technical implementation. Chapter 5 describes the evaluation of the proposed system, including the methodology, results. Chapter 6 discusses the implications of the findings and potential future work. Finally, Chapter 7 summarizes the current work and outlines future work.

Related Work

Text2Model Transformation? Text-based Process Modeling?

LLM-based Process Modeling Tool

ProMoAI

ProMoAI ProMoAI ...

BPMN-Chatbot

BPMN-Chatbot ...

AutoBPMN.AI

AutoBPMN.AI AutoBPMN.AI ...

Document AI Tool

NotebookLM?

...

Human-in-the-loop

Solution Design

Requirements

Functional Requirements

The system must meet the following core requirements for a MVP:

It shall integrate the necessary modeling functionalities within one unified interface (webpage).

It shall support upload multiple documents in textual format, e.g., TXT, DOCX, PDF.

It shall allow users to select multiple passages from one document to serve as input for LLM-based process model auto-generation.

It shall provide a clear visual link between the text and the generated model.

It shall embed the source text within the generated process model for traceability purposes.

It shall allow users to establish process-subprocess relationships between models.

It shall record user behavior and interactions with the system for analysis purposes.

One additional requirement is that the system shall be able to support document update.

Non-functional Requirements

Non-functional requirements for this tool mainly regard technical constraints:

It shall be implemented as a web-based application.

The frontend shall be developed using native web technologies (JavaScript, HTML, and CSS).

The visualization of the process model shall be implemented using the CPEE Layout¹ library.

The LLM-backed process model generation shall be implemented using the API from AutoBPMN.AI.

Design Overview

The design overview is described from the user perspective.

Firstly, we define the following concepts to better describe the system: - Project: a use case consisting of one or more documents for which process models needs to be created. - Workspace: the unified page/interface where the modeling workflow is performed, from document upload to model export.

¹<https://github.com/etm/cpee-layout>

- Traceability: the ability to navigate between source text and the corresponding process model. -
Project graph: a network-style visualization of relationships among models and documents within a project.

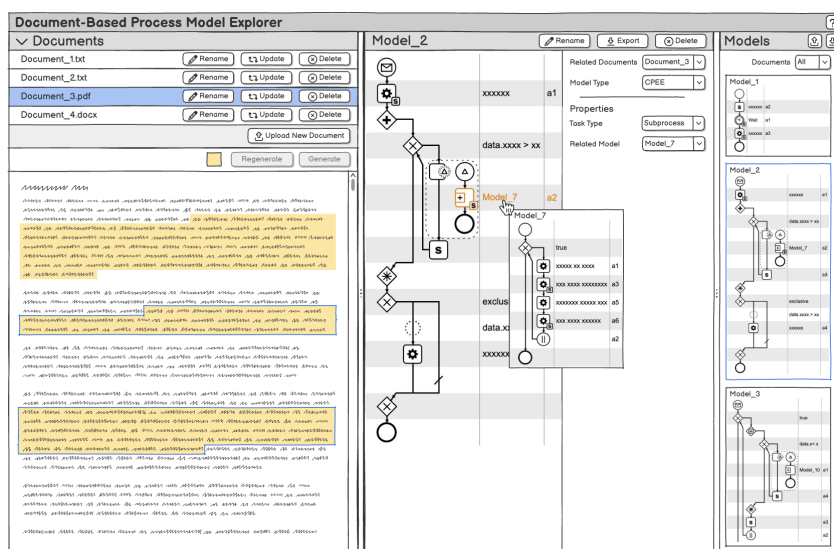
The system consists of three main pages: a home page for managing projects, a workspace page where the modeling workflow takes place, and a statistics page for viewing project-related metrics.

Views

The system is designed to contain three pages: a home page where a user can create a new project or open an existing project, a workspace page where the modeling workflow happens, and a statistic page where the user can view the statistics of the project.

Workspace. The layout design of the user interface is shown in the figure 1 in a 3-pane layout. The left pane is for document uploading and viewing, the middle pane is for process modeling and the right pane is for process models viewing and project graph visualization. The details are divided as the following: a) Document List b) Document Viewer c) Model Editor c.1) Model Canvas c.2) Model Prompt Editor d) Model List e) Project Graph Visualization.

Figure 1
Workspace UI Layout



The user interface layout of the workspace page.

Modeling Workflow Design

As the core part of the system, this section describes the modeling workflow that happens in the workspace page in Use Case Diagram. Figure: the use case diagram for the interaction of a user within the workspace page.

user → Upload/Rename/Update/Delete Document

→ View document → Select Text Passages → Generate Process Model

→ View Process Model → Modify Process Model → Link Process Model with other Models

→ Rename/Delete/Export Process Model

Core Functionalities Design

Document Management

Document Interaction

text selection text selection editing

Process modeling

Human-in-the-loop Process Modeling and Version Control.

Use Case: Model Generation. By click the button an advance mode is desgin

Use Case: Model Modification. a. Manual edit b. Update selected text and apply the changes. c. Refinement by sending prompt to LLM:

Input: prompt only and the current Model

d. Regeneration:

Input: type a. current model + changed selected text passage b. advanced mode: prompt + current model + changed selected text passage

Use Case: Establishing Process-Subprocess Relationships. - a review step for each LLM-generated model.

Figure 2*Syntax example of the augmented model data.*

```

<description xmlns:dbpm="...">
  <dbpm:info>
    <dbpm:document_info>
      <dbpm:document_id>...</dbpm:document_id>
      <dbpm:document_name>...</dbpm:document_name>
      <dbpm:document_version_id>...</dbpm:document_version_id>
      <dbpm:document_version_name>...</dbpm:document_version_name>
    </dbpm:document_info>
    <dbpm:text_selections>
      <dbpm:text_selection> ... </dbpm:text_selection>
      <dbpm:text_selection> ... </dbpm:text_selection>
    </dbpm:text_selections>
    <dbpm:prompts>
      <dbpm:prompt>... </dbpm:prompt>
      <dbpm:prompt>... </dbpm:prompt>
    </dbpm:prompts>
  </dbpm:info>
</description>...</description>
</description>

```

Augmented Model Data.***Traceability***

It is a

Project Graph Visualization

Visualize the

Interaction Design

The workspace support a highly freely interaction/navigation between model and documents.

A model can be triggered to be displayed by the following pathes:

- (1). Click on the model in the model list
- (2). Click on the text passage that is linked to the model
- (3). Click on the model tag displayed above the text passage
- (4). Click on the model node in the project graph visualization

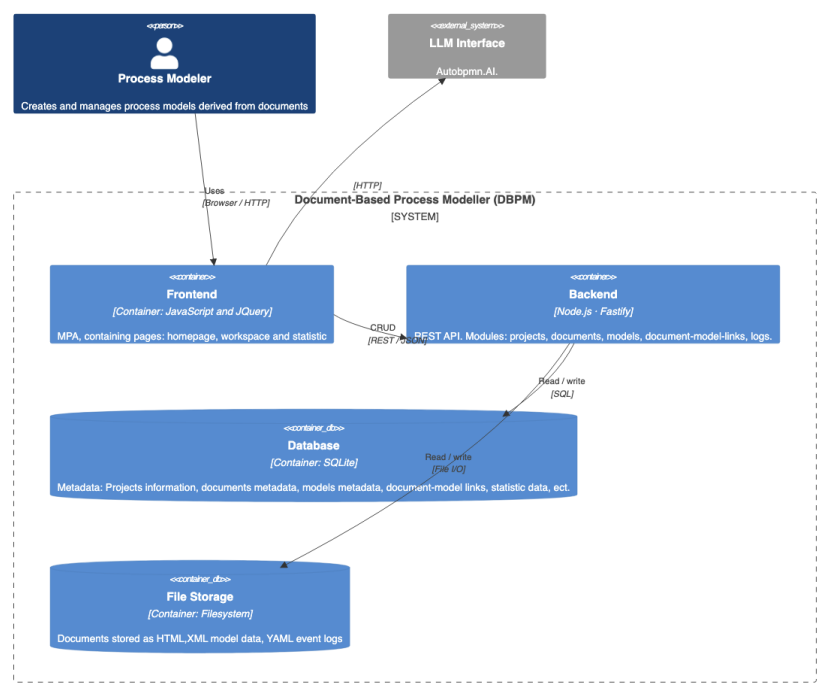
A document can be triggered to be displayed by the following pathes:

- (1). Click on the document in the document list
- (2). Click on the model in the model list, then its corresponding document will be displayed and the linked text passage's position will be auto-anchored.
- (3). Click on the document node in the project graph visualization

System Architecture

The system architecture is shown in the figure 3. The system is designed as a web-based application, with a front-end implemented using native JavaScript, HTML, and CSS. The back-end is responsible for handling user interactions, managing document uploads, and communicating with the LLM API for process model generation. The visualization of the process models is achieved using the CPEE library, which allows for dynamic rendering of BPMN diagrams based on the generated models.

Figure 3
System Architecture

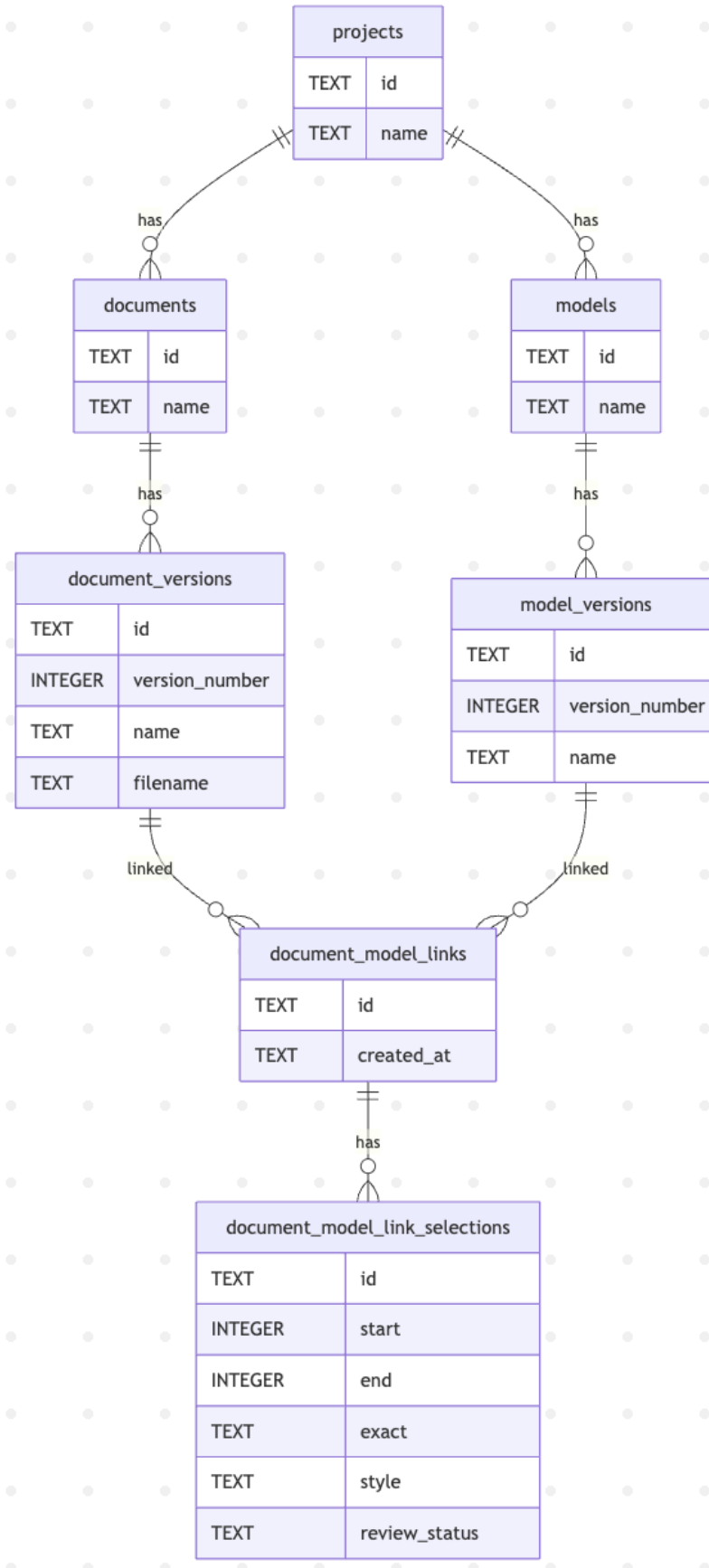


The system architecture of the document-based process modeling tool.

Conceptual Data Model

Figure 4

Conceptual Data Model



The system uses a TextQuoteSelector for robust anchoring, as defined in the W3C Web Annotation Model (Text Quote Selector section) **w3c_annotation_model**.

Implementation

Frontend

Architecture

Figure 5

Frontend codebase structure



Document Processing

Text Selection Rendering

Virtual Overlay Strategy.

Virtual Overlay Strategy.

Modeling

Rendering Customization

Architecture

Backend

Architecture

The tech-stack for the backend is currently chosen as Node.js with Fastify, as it provides out-of-the-box support for schema validation. As shown in the figure 6, the example code snippet of the backend

Figure 6

Example JSON schema used in the backend for text quotes

```
const textQuoteSchema = {  
  type: "object",  
  required: ["exact"],  
  additionalProperties: false,  
  properties: {  
    exact: { type: "string" },  
    prefix: { type: "string" },  
    suffix: { type: "string" },  
  },  
};
```

The backend implementation follows the layered Controller-Service-Repository architecture.

The backend component of the document-based process modeling tool.

Database

The second was a more realistic process modeling task based on an open-source SOP document ² in PDF format. For both tasks, participants were not given strict instructions on how to proceed, allowing them to explore the system freely. If time permitted, participants could additionally use the system with their own documents.

Following task completion, participants completed a questionnaire including the following questions:

- (1) What aspects of the tool did you like while using it?
- (2) What aspects of the tool did you not like or find problematic?
- (3) What improvements or additional features would you most like to see in the tool?
- (4) If you needed to model processes from documents in the future, would you consider using this tool? Why or why not?

Evaluation Results

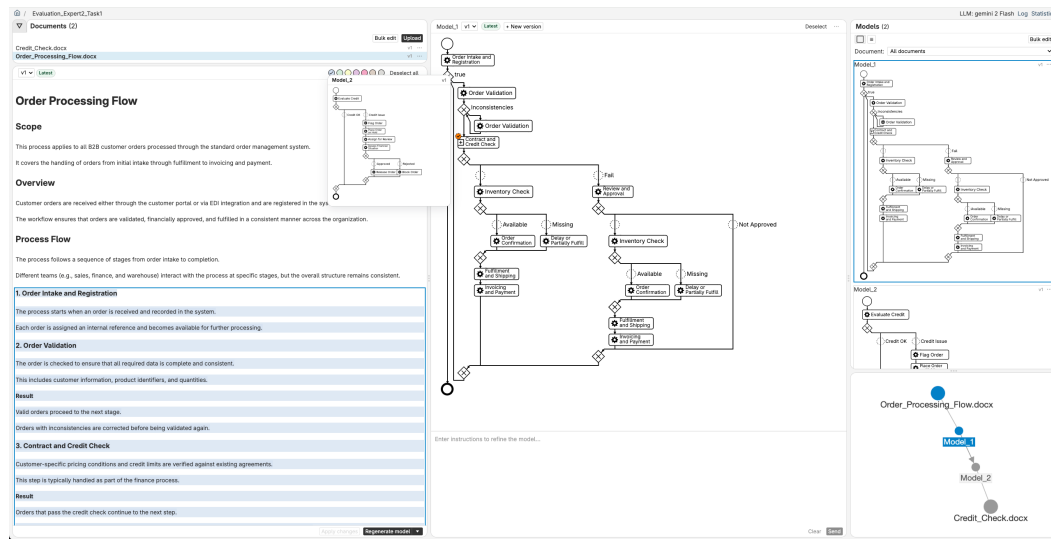
Overall Perception

Overall, participants perceived the tool as usable and intuitive, and capable of supporting document-based process modeling tasks within a unified environment. The Figure (See Figure 8). Figure 9 shows the modeling results from one expert for the first and second tasks, respectively.

The feedback to each functionality is summarized within the table 1. Due to different background of the expert, the perspectives varied. Details of the feedback are described in the following sections.

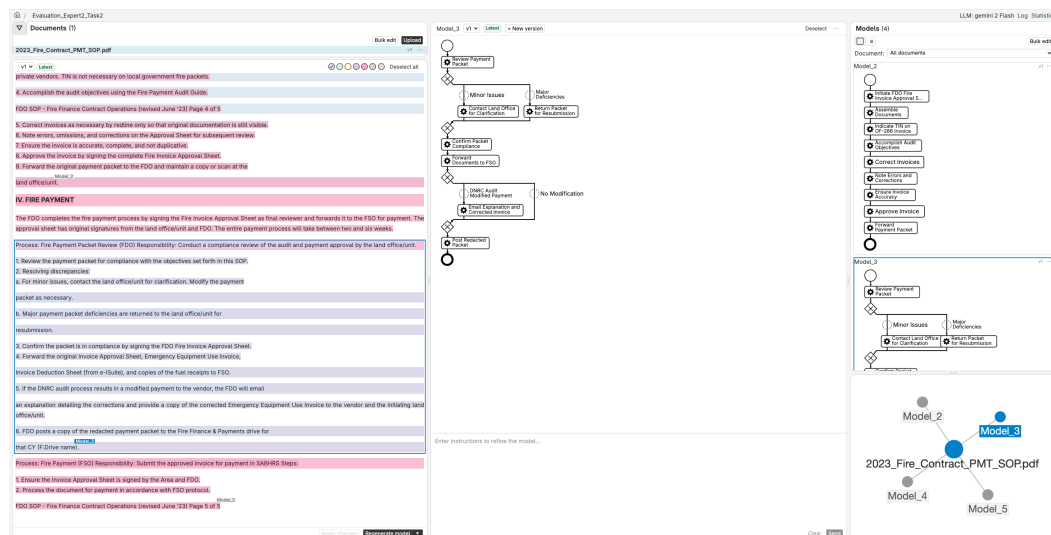
²DNRC SOP Document

Figure 8
Modeling result



Modeling result of an expert for the first task.

Figure 9
Modeling result



Modeling result of an expert for the second task.

Table 1
Summary of Evaluation Results by Functionality

Functionality		Positive Aspects		Challenges and Limitations	
		Item	n	Item	n
Document	Interac- tion			Readability issue	1
Integrated LLM-based Process Modeling		Fast	2	Unnecessary review step of the first generated model	2
		Good quality	2	Lack of transparency in regeneration logic	1
Augmented Data	Model	Well-considered	2		
Traceability		Clear	1	Lack of fine-grained link between actual used text and model	2
Process-Subprocess Relationship		Convenient	3		
Model Versioning		Helpful	3		
Project Graph		Helpful for larger systems	1	Node types not clear	1
Log and Statistics		Potentially useful for analysis	1		

n = number of participants who mentioned each item

Positive Aspects

Challenges in Interaction and Use

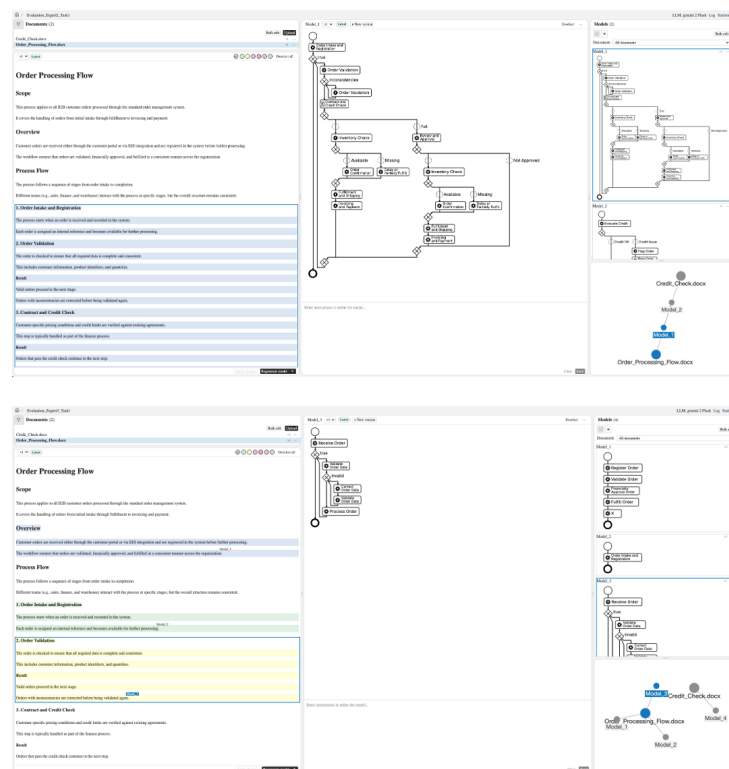
Potential Improvements and Additional Features

Additional Observations

Besides the gathered feedback, some findings regarding user behaviour and interaction are observed during the evaluation.

While most participants tended to select larger portions of text when starting with a new document, some preferred to begin with building models for smaller segments. These different strategies are illustrated in Figure 10.

Figure 10
Different modeling strategies



Different modeling strategies observed during task execution (above: larger part selection; below: step-by-step selection)

Discussion

This chapter discusses the findings of the evaluation and reflects on their implications for system design.

Usability and User Experience

There're some friction reported during the evaluation. While some features can be mitigated by design, some are hard. A this tool eliciate a new way of process modeling, it requires a little learnability to let user get used to the new way of modeling.

User Expectations

Conclusion

Summary

In this work, we propose and develop a process modeling tool, DBPM, with a series of convenient functionalities, especially for the case of process modeling from documents.

It provides a new modeling workflow for process modeling from documents, which allows users to upload documents, selects passages from the documents and auto-generate process models. The tool also maintains a clear link between the text and the generated model, and allows users to establish process-subprocess relationships between processes and subprocesses. We evaluate the tool with 4 experts from both academia and industry, and the results show that the tool is usable and intuitive, and capable of supporting document-based process modeling tasks within a unified environment. It provides a new interactive approach between the source text and the its corresponding model.

Future Work

Future work includes: - Improving document rendering - Fine-tuning the LLM for more precise text extraction

Appendix

Evaluation Material

B2B Order Processing Scenario

This use case consists of two synthetic documents.

Document 1: Order Processing Flow (excerpt)

Scope

This process applies to all B2B customer orders processed through the standard order management system. It covers the handling of orders from initial intake through fulfillment to invoicing and payment.

Overview

Customer orders are received either through the customer portal or via EDI integration and are registered in the system before further processing. The workflow ensures that orders are validated, financially approved, and fulfilled in a consistent manner across the organization. Process Flow The process follows a sequence of stages from order intake to completion. Different teams (e.g., sales, finance, and warehouse) interact with the process at specific stages, but the overall structure remains consistent.

1. Order Intake and Registration The process starts when an order is received and recorded in the system. Each order is assigned an internal reference and becomes available for further processing.
2. Order Validation The order is checked to ensure that all required data is complete and consistent. This includes customer information, product identifiers, and quantities. Result Valid orders proceed to the next stage. Orders with inconsistencies are corrected before being validated again.
3. Contract and Credit Check Customer-specific pricing conditions and credit limits are verified against existing agreements. This step is typically handled as part of the finance process. Result Orders that pass the credit check continue to the next step. Orders that fail the check are placed on hold for review and require approval before processing can continue.
4. Inventory Check The system verifies whether the requested items are available in stock. Result Orders with available items proceed to confirmation. Orders with missing items are delayed or partially fulfilled, depending on stock availability.
5. Order Confirmation Once the required checks are completed, the order is confirmed and released for execution. At this stage, the order is considered ready for

fulfillment. 6. Fulfillment and Shipping The warehouse prepares the order for shipment, including picking and packing. The goods are then shipped by the logistics provider. 7. Invoicing and Payment An invoice is generated and sent to the customer. Payment is processed according to the agreed terms. The process is considered complete once the invoice has been issued and payment is recorded in the system. Notes This document provides a high-level view of the order processing flow. Detailed rules and system-specific behavior are described in the corresponding process pages. Orders that are placed on hold remain in the process until the required issues have been resolved and approval has been granted. Related Pages For more detailed rules and step-specific behavior, see: - Order Validation - Credit Check

Document 2: Credit Check (excerpt) The credit check is performed after order validation and includes automated evaluation, issue detection, manual review, and decision-making.

Something Text here.

Next paragraph.